

	JERUSALEM COLLEGE OF ENGINEERING (An Autonomous Institution, Affiliated to Anna University, Chennai)
	QUESTION BANK

Subject code : JCS2201

Subject Name: PYTHON PROGRAMMING

Year / Semester : I/ II

Regulation : 2023

Branch : CSE,CS & BS,AIML,CS & IT

Unit No	Unit Name	Prepared by		
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Name :

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Date :

Academic Coordinator

Name :

Signature :

Date :

Head of the Department

Name :

Signature :

Date :

Note:

1. Verification to be done by senior faculty as allocated by the Department HOD.
2. Academic Coordinator and HOD of the department to which the faculty belongs must sign.
3. Verification for common subjects to be done by the Course Coordinator. In this case, Academic Coordinator and HOD of the department to which the course coordinator belongs must sign.

Knowledge Level as per Bloom's Taxonomy

K1- Remember; K2- Understand; K3- Apply; K4- Analyse; K5- Evaluate; K6- Create

K Level	Bloom's Taxonomy Action Verbs
K1	Choose, Define, Find, How, Label, List, Recall, Relate, Select, Show, Spell, What, When, Where, Which, Who, Why
K2	Classify, Compare, Contrast, Demonstrate, Explain, Extend, Illustrate, Infer, Interpret, Outline, Relate, Rephrase, Show, Summarize, Translate
K3	Apply, Build, Choose, Construct, Develop, Experiment with, Identify, Interview, Make use of, Model, Organize, Plan, Select, Solve, Utilize
K4	Analyze, Assume, Categorize, Classify, Compare, Conclusion, Contrast, Discover, Dissect, Distinguish, Divide, Examine, Function, Inference, Inspect
K5	Appraise, Assess, Award, Choose, Compare, Conclude, Criticize, Decide, Deduct, Defend, Determine, Disprove, Estimate, Evaluate, Explain, Importance, Influence, Interpret, Judge, Justify, Measure, Opinion, Perceive, Prioritize, Prove, Recommend
K6	Adapt, Build, Change, Choose, Combine, Compile, Compose, Construct, Create, Delete, Design, Develop, Discuss, Elaborate, Estimate, Formulate, Happen, Imagine, Improve, Invent, Makeup, Maximize, Minimize, Modify, Original, Originate, Plan, Predict, Propose, Solution, Solve, Suppose, Test, Theory

JERUSALEM COLLEGE OF ENGINEERING, CHENNAI 100
ACADEMIC YEAR EVEN (2024-2025)
JCS2201-PYTHON PROGRAMMING
I YEAR II SEM

UNIT-I INTRODUCTION TO PYTHON PROGRAMMING

Course objective: To provide an introduction to Python Programming Language.

Course Outcome: To understand the evolution of Python and run basic python programs.Taxonomy

Bloom's Taxonomy Level: K1,K2,K3,K4

PART A

Q.No.	Question	BTL*
1	What is Python? • Python is a general purpose, interpreted, open source, object- oriented, high level language. • Python is considered a scripting language and is often used for creating Web applications and dynamic Web content. • Python was developed by Guido van Rossum.	K1
2	What are the basic modes available in Python? • Python has two basic modes: ○ Normal/Interpreted/Script Mode ○ Interactive Mode • The Interpreted mode is the mode where the scripted and finished .py files are run in the Python interpreter. • Interactive mode is a command line shell which gives immediate feedback for each statement.	K1

3	What is the difference between intermediate mode and script mode? - In intermediate mode, you type Python expressions into the Python Interpreter window, and the interpreter immediately shows the result. For Example: <pre>>>> print('Keep calm and bubble on') >>> <i>Keep calm and bubble on</i> >>></pre> - Alternatively, you can write a program in a file and use the interpreter to execute the contents of the file. Such a file is called a script. Scripts have the advantage that they can be saved to disk, printed, and so on. For example: <pre>print(' This is a script mode window') print(' Exit from script mode")</pre> Using File Save and save the script under the name 'Test.py'. Execute the script from the Run menu or using F5. The above two messages will get printed. <pre><i>This is a script mode window</i> <i>Exit from script mode</i> >>></pre>	K2
4	List out the features of python. - Easy to Use - Expressive Language - Interpreted Language - Cross-platform language - Free and Open Source - Object-Oriented language - Extensible	K1

5	What are the different data types available in Python? The different data types available in Python are • Numbers (int, float, complex) • None • Bool or Boolean • List • Tuples • Dictionary • String • Set	K1
6	What is meant by python numbers? Number data types store numeric values. Number objects are created when you assign a value to them. Python supports four different numerical types: • int (signed integers) • long (long integers, they can also be represented in octal and hexadecimal) • float (floating point real values) • complex (complex numbers)	K1
7	What are python strings? • Strings in Python are identified as a contiguous set of characters represented in the quotation marks. • Python allows for either pairs of single or double quotes. • Subsets of strings can be taken using the slice operator ([] and [:]) with indexes starting at 0 in the beginning of the string and working their way from -1 at the end. • The plus (+) sign is the string concatenation operator and the asterisk (*) is the repetition operator. <u>Example:</u> <pre>str = 'Hello World!' print str[0] # Prints first character of the string</pre> <u>Output:</u> H	K1

8	Mention the features of lists in python • Lists are the most versatile of Python's compound data types. • A list contains items separated by commas and enclosed within square brackets []. • A list in Python is used to store the sequence of various types of data. • A list can be defined as a collection of values or items of different types. <u>For example,</u> <pre>list1 = [1, 2, "Python", "Program", 15.9]</pre> <pre>print(list1)</pre> <u>Output:</u> <pre>[1, 2, 'Python', 'Program', 15.9]</pre>	K1												
9	What is tuple? What is the difference between list and tuple? It is an absolute collection of Python objects separated by commas i.e a tuple can't be changed or modified after its creation. <table border="1"> <thead> <tr> <th>S.No</th><th>List</th><th>Tuple</th></tr> </thead> <tbody> <tr> <td>1</td><td>Lists are mutable</td><td>Tuples are immutable</td></tr> <tr> <td>2</td><td>The list is better for performing operations, such as insertion and deletion</td><td>Tuple data type is appropriate for accessing the elements.</td></tr> <tr> <td>3</td><td>Lists consume more memory</td><td>Tuple consumes less memory as compared to the list</td></tr> </tbody> </table>	S.No	List	Tuple	1	Lists are mutable	Tuples are immutable	2	The list is better for performing operations, such as insertion and deletion	Tuple data type is appropriate for accessing the elements.	3	Lists consume more memory	Tuple consumes less memory as compared to the list	K4
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10	What is a variable? • A variable is a name that refers to a value stored in memory location • Python variables are case sensitive. • Python variables can start with a letter a-z, A-Z or _(underscore), followed by letters, numbers or underscore. Ex: a=10, a is a variable	K1												
11	What are keywords? • Keywords are the reserved words in Python. • We cannot use a keyword as variable name, function name or any other identifier.	K1												

	• They are used to define the syntax and structure of the Python language • In Python, keywords are case sensitive. There are 33 keywords in Python. <u>Example:</u> False, class, finally, return	
12	What are the rules for writing an identifier? • Identifiers can be a combination of letters in lowercase (a to z) or uppercase (A to Z) or digits (0 to 9) or an underscore (_). Names like myClass, var_1 and print_this_to_screen, all are valid example. • An identifier cannot start with a digit. 1variable is invalid, but variable1 is perfectly fine. • Keywords cannot be used as identifiers. • We cannot use special symbols like!, @, #, \$, % etc. in our identifier. • Identifier can be of any length.	K1
13	How can we determine the data type of a given variable? • Everything in Python is an object, and every object has an identity, a type and a value. • To determine a variable type, we can use type () Ex: <pre>>>>a=456 >>>type(a) <class 'int'></pre>	K1
14	What are comments in python? • Comment tells the user/reader what a section of code does. • The comments in python begin with # symbol. • Comments are not read by interpreter and are non-executable statements <pre>#Addition of 2 numbers #c=a+b</pre>	K1

15	What is indent in Python? Why we require indentation in python? Python functions have no explicit begin or end, and no curly brace to mark where the function code starts and stops. The only delimiter is a colon (:) and the indentation of the code itself. Leading whitespace (spaces and tabs) at the beginning of a logical line is used to compute the indentation level of the line, which in turn is used to determine the grouping of statements.	K1
16	What are the rules in declaring Python variables? • A Python variable name must start with a letter or the underscore character. • A Python variable name cannot start with a number. • A Python variable name can only contain alpha-numeric characters and underscores (A-z, 0-9, and _). • Variable in Python names are case-sensitive (name, Name, and NAME are three different variables). • The reserved words (keywords) in Python cannot be used to name the variable in Python.	K1
17	What is Boolean value? The Boolean is a datatype that accepts two inputs. ○ True ○ False The value of True is 1 and False is 0	K1
18	What is a set in Python? A set is an unordered collection of items. Every element is unique (no duplicates) and must be immutable (which cannot be changed). However, the set itself is mutable. We can add or remove items from it. Sets can be used to perform mathematical set operations like union, intersection, symmetric difference etc. <u>Example:</u> <pre>#Create a Set thisset = {"apple", "banana", "cherry"} print(thisset)</pre> <u>Output:</u> <pre>{'cherry', 'banana', 'apple'}</pre>	K1

19	Can you change a set in Python? Sets are mutable. However, since they are unordered, indexing has no meaning. We cannot access or change an element of a set using indexing or slicing. Set data type does not support it.	K1
20	Define a keyword None. The None keyword is used to define a null value, or no value at all. None is not the same as 0, False, or an empty string. None is a data type of its own (None Type) and only None can be None. <u>Example:</u> <pre>x = None if x: print("Do you think None is True?") elif x is False: print("Do you think None is False?") else: print("None is not True, or False, None is just None...")</pre> <u>Output:</u> <i>None is not True, or False, None is just None...</i>	K1
21	Define operator precedence. Operators have different levels of precedence, which determine the order in which they are evaluated. When multiple operators are present in an expression, the ones with higher precedence are evaluated first.	K1
22	What is an expression? An expression is a combination of operators and operands that is interpreted to produce some other value. It has 8 types. 1. Constant Expressions 2. Arithmetic Expressions 3. Integral Expressions 4. Floating Expressions 5. Relational Expressions 6. Logical Expressions 7. Bitwise Expressions 8. Combinational Expressions	K1

23	Define Input and Print functions. In Python, program output is displayed using the print() function, while user input is obtained with the input() function. Python treats all input as strings by default, requiring explicit conversion for other data types. Example: <pre># Here, we take input from the user and then display it. name = input("Enter your first name: ") print("Hello! " + name)</pre>	K2
24	Define Command line arguments. The arguments that are given after the name of the program in the command line shell of the operating system are known as Command Line Arguments. Python provides various ways of dealing with these types of arguments. The three most common are: • <u>Using sys.argv</u> • <u>Using getopt module</u> • <u>Using argparse module</u>	K1

Q.No	PART – B	BTL*
1	Write a short note on history of Python.	K3
2	Illustrate interpreter and interactive mode in python with example.	K3
3	Write in detail about keywords in python with example.	K3
4	Write in detail about data types in python with suitable example.	K3
5	Explain in detail about the features of Python.	K3
6	Discuss the following data type conversions functions in detail with example. a) int() b) float() c) complex() d) bool()	K4
7	Write the following python programs. a. Exchange the value of two variable b. To find the sum of digits in a number c. To find the area and perimeter of a circle d. To find the square and cube of a number e. To find the roots of a quadratic equation f. To find the square root and cube root of a number g. To calculate Simple Interest h. To convert Celsius to Fahrenheit	K4
8	Explain Operators and its precedence with an example program.	K3
9	Explain Command line arguments.	K3

***BTL – Blooms Taxonomy Level**

UNIT II

CONTROL STRUCTURES AND STRINGS

Course Objective – To understand the control structures and string concepts.

Course Outcome – To develop program using control structure and simple string processing.

Bloom's Taxonomy Level – K1, K2, K3

PART A

S.No.	Question	BTL
1.	What are control structures? What are the different control structures used in Python? Control Structures are the blocks that analyze variables and choose directions in which to go based on given parameters. They can alter the control flow of a program execution. The primary types of control structures used in Python: - Sequential structure – default mode of execution - Selection structure - used for decisions and branching - Repetition structure - used for looping, i.e., repeating a piece of code multiple times	K1
2.	What is sequential structure? This control structure represents the linear and sequential execution of code, where statements are executed one after another in the order they appear. This is the simplest type of programming control structure and forms the foundation of most programs. Eg. <pre>a = 20 b = 10 c = a * b print(f"The product of {a} and {b} = {c}")</pre> OUTPUT The product of 20 and 10 = 200	K1
3.	What is selection structure? List the different selection control statements in Python. This control structure enables a program to choose between two or more paths of execution based on specific conditions. Several conditions are tested and instructions are executed based on which condition is true. The statements used for selection structure are also called as decision control statements or branching statements. They are - if statement - if-else statement - if – elif - else ladder statement	K1

	nested if statement	
4.	Write a python program that accepts two numbers, finds and displays the bigger one of them. <pre>A = int(input("Enter the first number")) B = int(input("Enter the second number")) if(A>B): print(A, "is bigger") else: print(B, "is bigger")</pre> <u>Output</u> Enter the first number80 Enter the second number90 90 is bigger	K3
5.	Write a Python program that allows a withdrawal transaction. Assume an available balance amount of Rs. 10,000. <pre>bal = 10000 amt = int(input("Enter the amount to withdraw: ")) if (amt >= bal): print("Insufficient balance. Withdrawal not possible.") else: bal -= amt print("Your balance amount = Rs. ", bal)</pre> <u>Output:</u> Enter the amount to withdraw: 8000 Your balance amount = Rs. 2000	K3
6.	What is iterative structure? List the different iteration statements in Python. This control structure allows certain blocks of code to be executed repeatedly as long as a condition remains true. Iteration statements in python include the following: - for loop - while loop	K2
7.	Write the syntax and usage of for loop. A for loop is a control flow statement that executes code repeatedly for a particular number of iterations. In this control flow statement, the keyword used is for. The 'for' loop is used when the number of iterations is already known. It has three forms of usage. (i) for var in range([start],end,[step]): statement block (ii) for var in list: statement block	K2

	(iii) for var in list: statement block 1 else: statement block 2 Usage: • When it is used with range() function the iterator variable var takes values from start_value to stop_value – 1. • When it is used with list / tuple, the iterator variable var takes values, one by one from the list / tuple. • Used to repeat a set of statements for a finite number of times. • Used to iterate through a string, list and tuple.	
8.	Write the syntax of while loop and its usage. It is a loop that executes a single statement or a group of statements for the given true condition. The keyword used to represent this loop is "while". A "while" loop is used when the number of iterations is unknown. It has two forms of usage. (i) while condition: statement block (ii) while condition: statement block 1 else: statement block 2 Usage: • Used to repeat a set of statements till a condition remains True. • Used to repeat a set of statements for a finite number of times. • Used to iterate through a string, list and tuple.	K1
9.	What is pass statement? ‘pass’ is a null statement. It is like a comment, but it is not ignored by the interpreter. When the user does not know what code to write, user simply places a ‘pass’ at that line. It can be placed where empty code is not allowed, like in loops, function definitions, class definitions, or in if statements. It can be used as a placeholder for future code. Syntax: pass	K2
10.	Can loop statements have else clause? If yes, when will it be executed? Yes. Python allows the else keyword to be used with the ‘for’ and ‘while’ loops too. The else block appears after the body of the loop. The statements in the else block will be executed after all iterations are completed. Eg. n = 4 for i in range(1, n): print(i)	K2

	else: print("PYTHON")											
11.	Differentiate between break and continue statements. <table><tr><th>Break</th><th>Continue</th></tr><tr><td>The break statement is used to terminate the execution of the nearest enclosing loop in which it appears.</td><td>The continue statement is used to stop the current iteration of the loop and continues with the next iteration</td></tr><tr><td>Syntax: break</td><td>Syntax: continue</td></tr><tr><td>It causes a forward jump in program execution.</td><td>It causes a backward jump in program execution.</td></tr><tr><td>Example i = 1 while(i <= 10): print(i, end="") if (i==5): break i=i+1 print("Done") Output: 1 2 3 4 5 Done</td><td>Example for i in range(1, 10): if (i % 5 == 0): continue print(i, end="") print("Done") Output: 1 2 3 4 6 7 8 9 Done</td></tr></table>	Break	Continue	The break statement is used to terminate the execution of the nearest enclosing loop in which it appears.	The continue statement is used to stop the current iteration of the loop and continues with the next iteration	Syntax: break	Syntax: continue	It causes a forward jump in program execution.	It causes a backward jump in program execution.	Example i = 1 while(i <= 10): print(i, end="") if (i==5): break i=i+1 print("Done") Output: 1 2 3 4 5 Done	Example for i in range(1, 10): if (i % 5 == 0): continue print(i, end="") print("Done") Output: 1 2 3 4 6 7 8 9 Done	K3
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12.	State the differences between continue and pass statement with an example. <table><tr><th>Continue</th><th>Pass</th></tr><tr><td>The continue statement is used to ignore the remaining statements in the current iteration of the loop and moves the control back to the start of the loop.</td><td>The pass statement is used when a statement is required syntactically.</td></tr><tr><td>It stops the current iteration.</td><td>It does nothing.</td></tr><tr><td>Syntax: continue</td><td>Syntax: pass</td></tr><tr><td>Example for i in range(1, 10): if (i % 5 == 0): continue print(i, end="") print("Done") Output: 1 2 3 4 6 7 8 9 Done</td><td>Example for i in range(1, 10): if (i % 5 == 0): pass print(i, end="") print("Done") Output: 1 2 3 4 6 7 8 9 Done</td></tr></table>	Continue	Pass	The continue statement is used to ignore the remaining statements in the current iteration of the loop and moves the control back to the start of the loop.	The pass statement is used when a statement is required syntactically.	It stops the current iteration.	It does nothing.	Syntax: continue	Syntax: pass	Example for i in range(1, 10): if (i % 5 == 0): continue print(i, end="") print("Done") Output: 1 2 3 4 6 7 8 9 Done	Example for i in range(1, 10): if (i % 5 == 0): pass print(i, end="") print("Done") Output: 1 2 3 4 6 7 8 9 Done	K3
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13.	Point out the errors in the following codes: (i) for i in range(1, 10, 2.0): print(i) (ii) while j <= 10: print(j) j++ (i) TypeError: 'float' object cannot be interpreted as an integer – float values cannot be used in range function. (ii) 2 errors: (a) SyntaxError: invalid syntax - ++ is not an operator in Python	K2
14.	Explain string comparison with an example. The comparison operator works on string to check if two strings are equal. <pre>>>>PIN='1234' >>>if PIN=='1234': print("Correct PIN") Correct PIN</pre>	K3
15.	What is meant by string slicing? Give examples. A substring of a string is obtained by taking a slice. The operator [n:m] returns the part of the string from the nth character to the mth character, including the first but excluding the last. Example: <pre>>>>book='Problem Solving and Python Programming' >>>print(book[0:7]) Problem >>>print(book[21:27]) Python</pre>	K2
16.	Mention the use of the split function in Python. The use of split function in Python is that it breaks a string into shorter strings using the defined separator. It gives a list of all words present in the string Syntax: string.split(separator, maxsplit) separator – Optional parameter - Specifies the separator to use when splitting the string. By default any whitespace is a separator maxsplit – Optional parameter - Specifies how many splits to do. Default value is -1, which is "all occurrences" Eg. <pre>txt = "hello world" x = txt.split(" ") print(x)</pre> Output: ['hello', 'world']	K2
17.	What is a Negative Indexing in Python? Negative Indexing is used to in Python to begin slicing from the end of the string i.e. the last. Slicing in Python gets a sub-string from a string. The slicing range is set as parameters i.e. start, stop, and step.	K2

	Eg. <pre>myStr = 'Python Programming!' print("String = ", myStr) print("String after slicing (negative indexing) = ", myStr[-4 : -1])</pre> Output: String after slicing (negative indexing) = ing																											
18.	What are the two operators that are used in string functions? The 2 operators used in string functions are ‘in’ and ‘not in’. They can be used to test whether a substring is a member of a string or not. <pre>>>>'t' in 'Python' True >>>'p' in 'Python' False >>>'thon' not in 'Python' False >>>'then' not in 'Python' True</pre>	K2																										
19.	What is the use of str.upper() and str.lower() functions in string?? The functions str.upper() and str.lower() will return a string with all the letters of original string converted to upper or lower case letters. <pre>>>>ss = 'Python Programming' >>>print(ss.upper()) PYTHON PROGRAMMING >>>print(ss.lower()) python programming</pre>	K2																										
20.	List few string methods and their purpose, used in Python. <table><thead><tr><th>Method</th><th>Description</th></tr></thead><tbody><tr><td>capitalize()</td><td>Converts the first character to upper case</td></tr><tr><td>find()</td><td>Searches the string for a specified value and returns the position of where it was found</td></tr><tr><td>index()</td><td>Searches the string for a specified value and returns the position of where it was found.</td></tr><tr><td>isalpha()</td><td>Returns True if all characters in the string are in the alphabet</td></tr><tr><td>isdigit()</td><td>Returns True if all characters in the string are digits</td></tr><tr><td>isnumeric()</td><td>Returns True if all characters in the string are numeric</td></tr><tr><td>islower()</td><td>Returns True if all characters in the string are lower case</td></tr><tr><td>isupper()</td><td>Returns True if all characters in the string are upper case</td></tr><tr><td>lower()</td><td>Converts a string into lower case</td></tr><tr><td>upper()</td><td>Converts a string into upper case</td></tr><tr><td>replace()</td><td>Returns a string where a specified value is replaced with a specified value</td></tr><tr><td>strip()</td><td>Returns a trimmed version of the string</td></tr></tbody></table>	Method	Description	capitalize()	Converts the first character to upper case	find()	Searches the string for a specified value and returns the position of where it was found	index()	Searches the string for a specified value and returns the position of where it was found.	isalpha()	Returns True if all characters in the string are in the alphabet	isdigit()	Returns True if all characters in the string are digits	isnumeric()	Returns True if all characters in the string are numeric	islower()	Returns True if all characters in the string are lower case	isupper()	Returns True if all characters in the string are upper case	lower()	Converts a string into lower case	upper()	Converts a string into upper case	replace()	Returns a string where a specified value is replaced with a specified value	strip()	Returns a trimmed version of the string	K3
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PART B

S. No	Question	BTL
1.	Explain in detail about the different conditional control statements / Selection Control statements / decision making with an example. (16 Marks)	K2
2.	What are the different looping construct available in python? Explain each of them in detail with relevant example. (16 Marks)	K2
3.	Write in detail about the different unconditional statements in Python. Give example program for each. (16 Marks)	K2
4.	a) Design a simple calculator using if ...elif construct. (16 Marks) b) Write a Python program to check the prime number using while loop.	K3
5.	a) Write a python program to print factorial of a given number (16 Marks) b) Implement a Python program to find the sum of n even natural numbers.	K3
6.	a) Develop a program to find the largest among three numbers. (16 Marks) b) Write a python program to find the given year is leap or not.	K3
7.	Discuss about the following operations related to strings in Python. (8 Marks) (a) String comparison (8 Marks) (b) String slicing (c) String splitting (d) String stripping	K2
8.	Explain the different string manipulation functions available in Python with example. (16 Marks)	K2

UNIT-III COLLECTIONS

Course objective: To learn different Python collections.

Course Outcome: create a code for different collection types and can differentiate them and to compose the code for functions, modules and packages.

Bloom's Taxonomy Level: K1,K2,K4

PART A

Q.No.	Question	BTL *
1	Define Python lists. Write major characteristics of lists. Python Lists - List is a collection of arbitrary objects, somewhat akin to an array in many other programming languages but more flexible. Lists are defined in Python by enclosing a comma-separated sequence of objects in square brackets ([]). The important characteristics of Python lists are as follows: ✓ Lists are ordered. ✓ Lists can contain any arbitrary objects. ✓ List elements can be accessed by index. ✓ Lists can be nested to arbitrary depth. ✓ Lists are mutable. ✓ Lists are dynamic	K1
2	What is list slicing? What the different ways of slicing? List slicing is considered a common practice in Python for accessing a range of elements from a list. The following is the different ways of slicing the list. Let Arr is the list. (i) Arr[start:stop]:- Print values from start to stop-1 (ii) Arr[start:]:- Print values from start to end (iii) Arr[:stop]:- Print values from beginning to stop-1 (iv) Arr[start:stop:step]:-Print values from start to stop-1 by skipping step	K1

3	. Print the output of the following code <pre>Birds=['Eagle', 'Sparrow', 'Crow', 'Parrot', 'Peacock'] print(Birds[:3]) print(Birds[-2:]) print(Birds[-2]) print(Birds[-1:-6:-1])</pre> Output <pre>['Eagle', 'Sparrow'] ['Parrot', 'Peacock'] ['Eagle', 'Sparrow', 'Crow'] ['Peacock', 'Parrot', 'Crow', 'Sparrow', 'Eagle']</pre>	K2
4	What is tuple? Tuple is an ordered sequence of items. It is defined within parentheses () where items are separated by commas. Tuples are immutable. Tuples once created cannot be modified. Tuples are used to write-protect data and are usually faster than list as it cannot change dynamically. <pre>>>> t = (5,'program', 34.66)</pre> We can use the slicing operator [] to extract items but we cannot change its value	K1
5	What are all the operations that can be performed in tuples? • len() - To find the length of the tuple. • Concatenation(+) - To concatenate two tuples. • Repetition(*) • Membership(in, not in)	K1
6	How to do tuple assignment? Give an example. Tuple assignment allows a tuple of variables on the left side of the assignment operator to be assigned values from a tuple given on the right side of the assignment operator. Each value is assigned to its respective variable. Tuple assignment occurs simultaneously rather than in sequence, making it useful for swapping values. Example: <pre>>>> tup=(100,200,300) >>> (val1,val2,val3)=tup</pre>	K1
7	What is tuple packing and tuple unpacking? In tuple packing, the values on the left are ‘packed’ together in a tuple: <pre>>>> b = ("Bob", 19, "CS") # tuple packing</pre> In tuple unpacking, the values in a tuple on the right are ‘unpacked’ into the variables/names on the right: <pre>>>> b = ("Bob", 19, "CS") >>> (name, age, studies) = b # tuple unpacking</pre>	K1

8	What are the advantages of tuple over list? • Tuples are immutable, iterating through tuples is faster than iterating over a list. This means that a tuple performs better than a list. • Tuples can be used as key for a dictionary but lists cannot be used as keys. • Tuples are best suited for storing data that is write-protected. • Tuples can be used in place of lists where the number of values is known and small. • If you are passing a tuple as an argument to a function, then the potential for unexpected behavior due to aliasing gets reduced. • Multiple values from a function can be returned using a tuple.	K1												
9	What is tuple? What is the difference between list and tuple? It is an absolute collection of Python objects separated by commas i.e a tuple can't be changed or modified after its creation. <table border="1"> <thead> <tr> <th>S.No</th><th>List</th><th>Tuple</th></tr> </thead> <tbody> <tr> <td>1</td><td>Lists are mutable</td><td>Tuples are immutable</td></tr> <tr> <td>2</td><td>The list is better for performing operations, such as insertion and deletion</td><td>Tuple data type is appropriate for accessing the elements.</td></tr> <tr> <td>3</td><td>Lists consume more memory</td><td>Tuple consumes less memory as compared to the list</td></tr> </tbody> </table>	S.No	List	Tuple	1	Lists are mutable	Tuples are immutable	2	The list is better for performing operations, such as insertion and deletion	Tuple data type is appropriate for accessing the elements.	3	Lists consume more memory	Tuple consumes less memory as compared to the list	K4
S.No	List	Tuple												
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10	What is list comprehension? Python also supports computed lists called list comprehensions having the following syntax. List = [expression for variable in sequence] Where, the expression is evaluated once, for every item in the sequence. • It is used to make new lists where each element is obtained by applying some operations to each member of another sequence or iterable. • List comprehension is also used to create a subsequence of those elements that satisfy a certain condition.	K1												
11	Define dictionary with an example. • Dictionaries are of compound type. They are Python's built-in mapping type. They map keys, which can be any immutable type, to values, which can be any type (heterogeneous), just like the elements of a list or tuple. • Dictionary is an unordered collection of key-value pairs. Each key is separated from its value by a colon (:), and consecutive items are separated by commas. The entire items in a	K1												

	dictionary are enclosed in curly brackets({}). The syntax for defining a dictionary is <pre>dictionary_name = {key_1: value_1, key_2: value_2, key_3: value_3}</pre> Example: dict1 = {'Name': 'Zara', 'Age': 7};	
12	List the methods that can be used on dictionaries. • Dict.get(key)- Returns the value for the key passed as argument. • Dict.clear()- Deletes all entries in dictionary • Dict.items()- Returns a list of key-value pairs. • Dict.keys()- Returns a list of keys in the dictionary. • len(dict)- Returns the length of the dictionary.	K1
13	Distinguish between lists and dictionaries. • First, a list is an ordered set of items. But, a dictionary is a data structure that is used for matching one item (key) with another (value). • Second, in lists, you can use indexing to access a particular item. But, these indexes should be a number. In dictionaries, you can use any type (immutable) of value as an index. • Third, lists are used to look up a value whereas a dictionary is used to take one value and look up another value. For this reason, dictionary is also known as a lookup table. • Fourth, the key-value pair may not be displayed in the order in which it was specified while defining the dictionary. This is because Python uses complex algorithms (called hashing) to provide fast access to the items stored in the dictionary. This also makes dictionary preferable to use over a list of tuples.	K1
14	What are the properties of dictionary keys? • More than one entry per key is not allowed (no duplicate key is allowed) • The values in the dictionary can be of any type while the keys must be immutable like numbers, tuples or strings. • Dictionary keys are case sensitive- Same key name but with the different case are treated as different keys in Python dictionaries.	K1
15	What is traversing in list? give example Traversing a list means accessing and processing each of the elements in a list(visiting all the elements once). For loop is mainly used to traverse a list. <i>Syntax:</i> <pre>for variable_name in list_name: Process each item Print or operations</pre>	K1

16	What are dictionary functions and methods? Python dictionary mainly has 8 inbuilt functions & methods. They are: • len() • clear() • get() • has_key() • items() • keys() • values() • cmp()	K1
18	What is a set in Python? A set is an unordered collection of items. Every element is unique (no duplicates) and must be immutable (which cannot be changed). However, the set itself is mutable. We can add or remove items from it. Sets can be used to perform mathematical set operations like union, intersection, symmetric difference etc. <u>Example:</u> <pre>#Create a Set thisset = {"apple", "banana", "cherry"} print(thisset)</pre> <u>Output:</u> <pre>{'cherry', 'banana', 'apple'}</pre>	K1
19	Can you change a set in Python? Sets are mutable. However, since they are unordered, indexing has no meaning. We cannot access or change an element of a set using indexing or slicing. Set data type does not support it.	K1

20	What are the operations that can be performed in Set? The None keyword is used to define a null value, or no value at all. None is not the same as 0, False, or an empty string. None is a data type of its own (None Type) and only None can be None. <u>Example:</u> <pre> x = None if x: print("Do you think None is True?") elif x is False: print("Do you think None is False?") else: print("None is not True, or False, None is just None...") </pre> <u>Output:</u> <i>None is not True, or False, None is just None...</i>	K1
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Q.No	PART – B	BTL*
1	Illustrate the following list methods with syntax and example. Get input from the user and write the relevant output(16) - Append - remove - pop - insert - count - index - reverse - sort	K3
2	Do the following operations on the list. (8 marks) - Merge two list - Convert string, tuple and set into list - Create a shallow copy - Clone a list	K3
3	Explain tuple operations and methods in detail with neat example(16)	K3

4	Define Dictionary in Python and explain all operations and methods in detail with example programs(16)	K3
5	What is tuple in python? How does it differ from list?.Explain in detail with example	K3
6	Given the following dictionary: portfolio = { 'accounts' : ['SBI', 'IOB'], 'shares' : [HDFC, 'ICICI', 'TM', 'TCS'], 'ornaments' : ['10 gm gold', '1 kg silver'] } Write a program to perform the following operations: - Add a key to portfolio called 'MF' with values 'Relaince' and 'ABSL'. - Set the value of 'accounts' to a list containing 'Axis' and 'BOB'. - Sort the items in the list stored under the 'shares' key. - Delete the list stored under 'ornaments' key.	K4

***BTL – Blooms Taxonomy Level**

UNIT IV

FUNCTIONS, MODULES AND PACKAGES

Course objective: To acquire knowledge on functions, modules and packages

Course Outcome: learn how to handle function and create modules.

Bloom's Taxonomy Level – K1, K2, K3, K4

PART A

Q.No.	Question	BTL*
1	Define function and write main the advantages of function. A function is a block of organized, reusable code that is used to perform a single, related action. Functions provide better modularity for your application and a high degree of code reusing. Two main advantages of functions are: (a) They help us to divide our program into multiple tasks. For each task we can define a function. This makes the code modular. (b) Functions provide a reuse mechanism. The same function can be called any number of times.	K1
2	What are the types functions with example? (a) Built-in functions Built-in functions are designed & developed by software provider. We can only able to use not able to modify them. Example. len(), sorted(), min(), max() etc... (b) User-defined functions. User Defined Functions are the functions defined by user to perform specific task. Function Implementation should starts with a keyword def , It introduces function definition.	K1
3	What is scope of variables? Scope of a variable is the region in the code where the variable is available/accessible. A variable declared outside a function (i.e. the main region of the code) is called a global variable and a variable declared inside a function is called a local variable of that function.	K1
4	Difference between parameter argument. A parameter is the variable listed inside the parentheses in the function definition. An argument is the value that are sent to the function when it is called.	K1

5	What is arbitrary function? Python Arbitrary Arguments allows a function to accept any number of positional arguments i.e. arguments that are non-keyword arguments, variable-length argument list. Example – Copy Code.	K1
6	What is arbitrary list in Python? An arbitrary argument list is a Python feature to call a function with an arbitrary number of arguments. It's based on the asterisk “unpacking” operator *. To catch an arbitrary number of function arguments in a tuple args, use the asterisk syntax *args within your function definition.	K1
7	What is Recursive Functions ? Recursion is the process of defining something in terms of itself. This means that the function will continue to call itself and repeat its behavior until some condition is met to return a result.	K1
8	What is Lambda function? A lambda function is a small anonymous function. A lambda function can take any number of arguments, but can only have one expression. Syntax: lambda arguments : expression	K1
9	What is module and package in Python? In Python, module is the way to structure program. Each Python program file is a module, which imports other modules like objects and attributes	K1
10	How to create a date object in python?. <pre>import datetime x = datetime.datetime(2020, 5, 17) print(x)</pre> O/p: 2020-05-17 00:00:00	K3
11	What is the output of the below code: <pre>def my_function(fname): print(fname + " in_JCE") my_function("CSE") my_function("CSBS")</pre>	K3

	<pre>my_function("AIML") my_function("CS") my_function("IT")</pre>	
12	What is the output of the below code: <pre>x = lambda a: a + 10 print(x(5))</pre>	K3
13	What is Direct Recursion? Direct recursion occurs when a function calls itself directly. It can be categorized into four subtypes: tail recursion, head recursion, tree recursion, and nested recursion.	K1
14	What is Tail Recursion? This type of recursion is a type of direct recursion in which the recursive call is the last operation in the function. This allows the compiler or interpreter to optimize the recursion, as it doesn't need to maintain a stack of function calls.	K1
15	Define strftime(). The strftime function, which stands for 'string format time', is part of Python's datetime module. It's used to convert date and time objects into strings, allowing you to present them in a more human-readable format.	K2
16	What is the output of the below code: <pre>x = min(5, 10, 25) y = max(5, 10, 25) print(x) print(y)</pre>	K3
18	What is the use of the below methods <pre>abs() pow(x, y) math.ceil() math.floor()</pre>	K2

19	How to create a module. To create a module just save the code you want in a file with the file extension .py: Example: Save this code in a file named mymodule.py <pre>def greeting(name): print("Hello, " + name)</pre> Now we can use the module we just created, by using the import statement: <pre>import mymodule mymodule.greeting("JCE")</pre>	K1
20	How to import module from a package? In Python, we can import modules from packages using the dot (.) operator.	K1

Q.No	PART – B	BTL*
1	Explain function and its types with syntax and example.	K2
2	Write about the argument types with syntax and example	K2
3	Explain scope of variable in python.	K2
4	Write a python program to find the factorial of given number using recursive function.	K3
5	What is lambda function and explain with example.	K2
6	Explain date and time methods in python.	K4
7	Explain modules and packages in python	K3

***BTL – Blooms Taxonomy Level**

UNIT V

FILES, EXCEPTIONS AND GUI

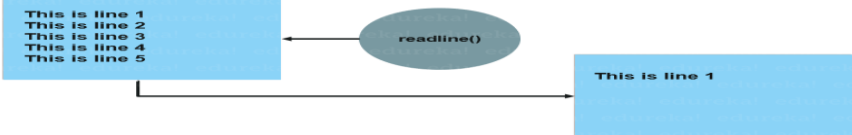
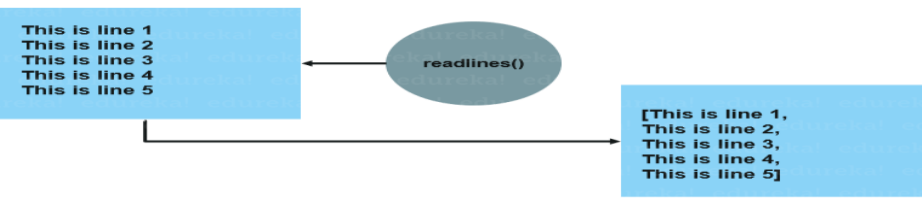
Course objective: To comprehend the file and exception handling concepts and GUI

Course Outcome: learn to handle file and exception and to design user interface using Tkinter.

Bloom's Taxonomy Level – K1, K2, K3, K4

PART A

Q.No.	Question	BTL*
1	Define File. List out its types. Files are named locations on disk to store related information. They are used to permanently store data in a non-volatile memory (e.g. hard disk). File types: • Binary File • Text File	K1
2	What is the difference is between append and write mode? Append in a Python File: To append data into a file open the file in 'a' mode. Append mode writes data at the end of the file Example: <pre>m = open("C:/Documents/Python/test.txt", "a+") m.write("Strawberry")</pre> The above code appends the string 'Strawberry' at the end of the 'test.txt' file Write mode: is used to write from the beginning of a file. If the file is an existing file, write mode will overwrite the information on the existing file. Example: <pre>f = open("C:/Documents/Python/test.txt", "w") f.write("Hello World")</pre> The above code writes the String 'Hello World' into the 'test.txt' file.	K1
3	Explain the moving methods of turtle? There are four directions that a turtle can move in: • Forward • Backward • Left • Right Initialize the variable t, to refer the turtle: <pre>>>> t = turtle.Turtle() >>> t.right(90) >>> t.forward(100) >>> t.left(90) >>> t.backward(100)</pre>	K2
4	Explain open() and close() methods for opening and closing a file. The open() Method. The open() function opens a file, and returns it as a file object. Syntax: <pre>open(file, mode)</pre> The close() Method The close() method of a file object flushes any unwritten information and closes the file object, after which no more writing can be done. Python automatically closes a file when the reference object of a file is reassigned to another file. Syntax: <pre>fileObject.close()</pre>	K2

5	What are additional attributes of turtle? 1. Turtle Visibility - Methods <code>showturtle()</code> and <code>hideturtle()</code> control a turtle's visibility. 2. Turtle Size- The size of a given turtle shape can be controlled with methods <code>resizemode()</code> and <code>turtlesize()</code>. 3. Turtle Shape- A turtle's shape may be set to one of the provided shapes, a described polygon (or collection of polygons), or an image. 4. Turtle Speed- The speed of a turtle can be controlled by use of the speed method.	K1
6	State the use of <code>read()</code> and <code>readline()</code> function in python file handling mechanism. Python <code>readline()</code> method will return a line from the <u>file</u> when called.  <code>readlines()</code> method will return all the lines in a file in the format of a <u>list</u> where each element is a line in the file. 	K1
7	Write a Python program to display a square with color fill. <pre> import turtle t = turtle.Turtle() t.fillcolor('red') t.begin_fill() for _ in range(4): t.forward(100) t.right(90) t.end_fill() </pre>	K1
8	What are various packages available for Python graphics and GUI? - Turtle - Tkinter - Open CV - PyGUI - PyForms - OPENGL - wxPython etc..	K1
9	How to open a new file in Python? Opening a file creates a file object. In this example, the variable <code>f</code> refers to the new file object. <pre> >>> f = open("test.dat","w") >>> print f </pre> The <code>open</code> function takes two arguments. The first is the name of the file, and the second is the mode. Mode " <code>w</code> " means that we are opening the file for writing	K1
10	Explain how the write method works on a file. <pre> >>> f.write("Now is the time") >>> f.write("to close the file") >>> f.close() </pre>	K2

11	Which method is used to read the contents of a file which is already created? The read method reads data from the file. With no arguments, it reads the entire contents of the file: <pre>>>> f = open("test.dat","r") >>> text = f.read() >>> print(text) >>> f.close()</pre>	K1										
12	What is a text file? Give an example for a text file. A text file is a file that contains printable characters and whitespace, organized into lines separated by newline characters. Example : To create a text file with three lines of text separated by newlines: <pre>>>> f = open("test.dat","w") >>> f.write("line one\n line two\n line three\n") >>> f.close()</pre>	K1										
13	What is an exception? Explain with few examples. Whenever a runtime error occurs, it creates an exception. Usually, the program stops and Python prints an error message. Example: Example 1. <pre>>>>3/0</pre> Name of Error will be ZeroDivisionError And the reason for the error is division by zero Example 2 <pre>>>> list1=[2, 3, 4, 5, 6] >>>list1[5]</pre> Name of Error is :IndexError Reason of Error is : list index out of range Example 3. >>>x+3 Name of Error is : Name Error Reason of Error is : name 'x' is not defined" Example 4. >>>int('22.5') Name of Error is : ValueError Reason of error is : invalid literal for int() with base 10: '22.5' Example 5. >>> '2'*'3' Error name is :TypeError	K1										
14	What are various types of errors? Compilation errors / syntax errors Logical errors (incorrect results due to wrong code) Run time errors/ Exceptions Built-in exceptions (Provided by Python eg KeyError, IOError etc) User defined exceptions / custom exceptions	K1										
15	What are the keywords related to exception handling? <table><tr><td>try</td><td>Code to be executed is coded in try block</td></tr><tr><td>except</td><td>error handling code</td></tr><tr><td>else</td><td>code to execute if no errors</td></tr><tr><td>finally</td><td>this code runs in any case (error / no error)</td></tr><tr><td>raise</td><td>throw exception</td></tr></table>	try	Code to be executed is coded in try block	except	error handling code	else	code to execute if no errors	finally	this code runs in any case (error / no error)	raise	throw exception	K1
try	Code to be executed is coded in try block											
except	error handling code											
else	code to execute if no errors											
finally	this code runs in any case (error / no error)											
raise	throw exception											
16	What are the error messages that are displayed for the following exceptions? a. Accessing a non-existent list item b. Accessing a key that isn't in the dictionary c. Trying to open a non-existent file IndexError: list index out of range KeyError IOError No such file or directory: 'filename'	K1										
17	Explain briefly about Python turtle. turtle is a pre-installed Python library that enables users to create various shapes by providing with a virtual canvas. The onscreen pen that is used for drawing on a screen is called the turtle and this is what gives the library its name. Various shapes can be created	K2										

	by moving turtle in different directions. Eg turtle.penup() , turtle.goto(200,0) , turtle.pendown(), turtle.forward(50) turtle.left(100) etc	
18	What is Turtle Graphics? > Turtle graphics refers to a means of controlling a graphical entity (a “turtle”) in a graphics window with x, y coordinates. > Python provides the capability of turtle graphics in the turtle Python standard library module. > There may be more than one turtle on the screen at once. Each turtle is represented by a distinct object.	K1
19	What are the fundamental turtle attributes? > Turtle objects have three fundamental attributes: 1. position 2. heading (orientation) 3. pen attributes.	K1
20	What is Tkinter module in Python? Tkinter module in graphics programming is the process of creating graphical interfaces by providing a set of methods and classes for building windows, buttons, labels, entry fields, and more. It allows developers to create sophisticated applications.	K1

Q.No	PART – B	BTL*
1	What are the two types of files? Explain different file operations (16)	K1
2	Differentiate Errors and Exceptions. (8) Illustrate try-except-else. (8)	K4
3	a) Explain about the different types of Exceptions in Python. (6 marks) b) Describe about Handling Exceptions in detail with examples. (10 marks)	K2
4	Explain in detail about Python Files, its types, functions and operations that can be performed on files with examples. (16 marks)	K2
5	Write a python program using turtle to draw different 2D drawing shapes and color them. (16)	K1
6	Explain the GUI toolkit Tkinter. (16)	K2
7	Design a Simple registration form using Tkinter in Python (16)	K3

***BTL – Blooms Taxonomy Level**